A sharp weak-driving phase bound for symmetric complete Markov networks

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Abstract

Periodically driven Markov networks generally respond with a phase lag. For a complete N-state network with a common attempt rate and symmetric splitting of energy differences, we prove a sharp bound on the leading weak-driving response. For every $N \geq 3$, every finite vector of energy offsets, and every finite real Fourier drive, the quadratic response coefficients obey $A_2 \geq 2k_0B_2$. The proof first expresses A_2/B_2 as a positive weighted average of the nonzero relaxation rates. The symmetrized generator is then reduced to the rank-one matrix $\mathrm{diag}(1/q_i) - \mathbf{1}\mathbf{1}^{\mathsf{T}}$, whose smallest positive eigenvalue is shown to be strictly larger than 2 for every positive probability vector q when $N \geq 3$. We also classify equality: at finite offsets it occurs only for zero-response drives, whereas every drive that changes an energy difference gives a strict inequality. An explicit two-heavy-state family approaches the bound as the remaining stationary weights vanish, proving that the constant 2 is optimal but unattained by a response-active finite-offset system. The result is confined to the weak-amplitude, equal-prefactor, symmetric-splitting complete graph; broader finite-amplitude and heterogeneous-network bounds remain open.

Keywords: stochastic thermodynamics; periodic driving; phase lag; Markov networks; spectral gap; rank-one perturbation

1 Introduction

A periodic modulation of a stochastic system is followed by a delayed response. For finite-state Markov dynamics, Degünther, Koyuk and Seifert introduced a generalized phase that remains meaningful for multi-harmonic driving and related it to cycle-averaged contractions of the probability distribution with the first and second time derivatives of the energies [1]. They established a tight two-state bound. For multi-state networks with a common rate amplitude they reported numerical evidence and left the corresponding general proof open; their heterogeneous-amplitude statement is broader still.

Here we resolve a sharply delimited part of that problem. We consider the complete graph K_N, a common attempt rate k_0, symmetric splitting, finite energy offsets, and the leading nonvanishing order in the driving amplitude. The response problem then becomes a spectral-gap problem for a nonuniform diagonal matrix minus a rank-one matrix. Rank-one perturbations are a standard tool in graph spectral theory [2], but the offset-dependent matrix here has kernel proportional to a nonuniform vector, and immediate interlacing does not settle the regime in which a single auxiliary weight exceeds one half.

Our main result is

$$A_2 \geq 2k_0B_2. \tag{1}$$

For $N \geq 3$ and finite offsets, the inequality is strict whenever the drive changes at least one energy difference. Equality is precisely the zero-response class. The constant is nevertheless sharp because a family with two equally weighted dominant states converges to the symmetric

two-state mechanism. Besides proving (1), this identifies why adding finitely occupied states cannot increase the weak-drive lag beyond that limiting mechanism.

Section 2 defines the physical domain and derives the modal response. Section 3 proves the exact spectral inequality. Section 4 classifies equality and establishes sharpness. We close with the physical scope and the unresolved extensions.

2 Model and weak response

2.1 Complete detailed-balance network

Let $N \geq 3$, $k_0 > 0$, and $T > 0$, with fundamental frequency $\Omega = 2\pi/T$. Energies are measured in units in which $k_{\mathrm{B}} T_{bath} = 1$. The rate from state i to state j is

$$k_{ij}^{(\varepsilon)}(t) = k_0 \exp\left[\frac{E_i^{(\varepsilon)}(t) - E_j^{(\varepsilon)}(t)}{2}\right], \qquad E_i^{(\varepsilon)}(t) = E_i^0 + \varepsilon e_i(t), \tag{2}$$

where all E_i^0 are finite and each e_i is a finite real Fourier series of period T. With $p(t)$ a normalized column vector, the master equation is

$$\dot{p}_i(t) = \sum_{j \neq i} \left[p_j(t) k_{ji}(t) - p_i(t) k_{ij}(t)\right]. \tag{3}$$

All rates are positive, so the long-time T-periodic state is unique.

For that periodic state, define

$$\mathcal{A}(\varepsilon) = \frac{1}{T}\int_0^T p^{(\varepsilon)}(t) \cdot \ddot{E}^{(\varepsilon)}(t)\, dt, \qquad \mathcal{B}(\varepsilon) = \frac{1}{T}\int_0^T p^{(\varepsilon)}(t) \cdot \dot{E}^{(\varepsilon)}(t)\, dt. \tag{4}$$

The constant and linear terms vanish by periodicity. We use the coefficient convention

$$A_2 = \lim_{\varepsilon \to 0} \frac{\mathcal{A}(\varepsilon)}{\varepsilon^2}, \qquad B_2 = \lim_{\varepsilon \to 0} \frac{\mathcal{B}(\varepsilon)}{\varepsilon^2}. \tag{5}$$

Thus A_2 has units of inverse time squared, B_2 has units of inverse time, and A_2/B_2 is a relaxation rate when $B_2 > 0$.

At $\varepsilon = 0$, introduce

$$x_i = e^{-E_i^0/2}, \qquad S = \sum_i x_i, \qquad Q = \sum_i x_i^2, \qquad \pi_i = \frac{x_i^2}{Q}. \tag{6}$$

The equilibrium vector is π. If L_0 denotes the column-conservative generator in (3) and $\Pi = \mathrm{diag}(\pi_i)$, detailed balance gives the real symmetric, symmetrized negative generator

$$M = -\Pi^{-1/2} L_0 \Pi^{1/2} = k_0 \left[S\, \mathrm{diag}(x_1^{-1}, \ldots, x_N^{-1}) - \mathbf{1}\mathbf{1}^{\mathsf{T}}\right], \qquad Mx = 0. \tag{7}$$

2.2 Spectral representation of the response

Write the first-order periodic response as $p^{(\varepsilon)} = \pi + \varepsilon r + O(\varepsilon^2)$ and set $r = \Pi^{1/2} z$. For

$$\widetilde{e}(t) = e(t) - \mathbf{1}\, \pi^{\mathsf{T}} e(t), \qquad h(t) = \Pi^{1/2} \widetilde{e}(t), \tag{8}$$

The tangent to the instantaneous equilibrium distribution is $\delta\pi = -\Pi\widetilde{e}$. Differentiating $L(E)\pi(E) = 0$ therefore gives $L'[e]\pi = L_0 \Pi \widetilde{e}$. Substitution in the first-order master equation $\dot{r} = L_0 r + L'[e]\pi$, followed by the similarity transform in (7), gives

$$\dot{z} = -Mz - Mh. \tag{9}$$

Normalization makes z orthogonal to $\ker M$, so only the positive modes enter.

To make all Fourier factors explicit, write

$$e(t) = e^{(0)} + \sum_{n=1}^{H} \mathrm{Re}\left[c_n e^{\mathrm{i}\omega_n t}\right], \qquad \omega_n = n\Omega, \tag{10}$$

and let $h_n = \Pi^{1/2}(c_n - \mathbf{1}\pi^{\mathsf{T}} c_n)$. Choose an orthonormal positive-mode eigenbasis $Mu_\mu = \lambda_\mu u_\mu$, and put $a_{n\mu} = u_\mu^{\mathsf{T}} h_n$.

Proposition 1 (Modal response)**.** *For the model* (2)–(10)*,*

$$A_2 = \frac{1}{2} \sum_{n,\mu>0} \frac{\omega_n^2 \lambda_\mu^2 |a_{n\mu}|^2}{\lambda_\mu^2 + \omega_n^2}, \tag{11}$$

$$B_2 = \frac{1}{2} \sum_{n,\mu>0} \frac{\omega_n^2 \lambda_\mu |a_{n\mu}|^2}{\lambda_\mu^2 + \omega_n^2}. \tag{12}$$

Consequently, when $B_2 > 0$, A_2/B_2 is a positive weighted average of the nonzero eigenvalues of M.

Proof. For a mode with frequency ω_n and eigenvalue λ_μ, (9) gives the complex response amplitude

$$Z_{n\mu} = -\frac{\lambda_\mu}{\lambda_\mu + \mathrm{i}\omega_n} a_{n\mu}. \tag{13}$$

Expanding (4) to order ε^2 yields $A_2 = \langle r \cdot \ddot{e} \rangle_T$ and $B_2 = \langle r \cdot \dot{e} \rangle_T$. Because z is orthogonal to $\Pi^{1/2}\mathbf{1}$, these contractions equal $\left\langle z \cdot \ddot{h} \right\rangle_T$ and $\left\langle z \cdot \dot{h} \right\rangle_T$. Taking the real-period average of each complex harmonic in (13) gives (11) and (12). Every weight in the latter is nonnegative and is positive exactly when the corresponding $a_{n\mu}$ is nonzero. □

For $N = 2$ at equal offsets, the sole positive eigenvalue is $2k_0$, recovering $A_2 = 2k_0 B_2$ and the symmetric two-state phase reported in Ref. [1]. The remaining task is therefore an all-N lower bound on the positive spectrum of (7).

3 A sharp spectral gap

Normalize x in the one-norm by defining $q_i = x_i/S$. Then $q_i > 0$, $\sum_i q_i = 1$, and

$$\frac{M}{k_0} = A(q), \qquad A(q) = \mathrm{diag}(q_1^{-1}, \ldots, q_N^{-1}) - \mathbf{1}\mathbf{1}^{\mathsf{T}}. \tag{14}$$

Lemma 2 (Positivity and kernel)**.** *$A(q)$ is positive semidefinite and* $\ker A(q) = \mathrm{span}\{q\}$.

Proof. For $y \in \mathbb{R}^N$, put $b_i = y_i/q_i$. Then

$$y^{\mathsf{T}} A(q) y = \sum_i q_i b_i^2 - \left(\sum_i q_i b_i\right)^2 \tag{15}$$

$$= \sum_{i<j} q_i q_j (b_i - b_j)^2 \geq 0. \tag{16}$$

Since every q_i is positive, equality holds exactly when all b_i are equal, or equivalently when y is proportional to q. □

Theorem 3 (Strict finite-offset gap)**.** *For every $N \geq 3$ and every positive vector q with $\sum_i q_i = 1$,*

$$\lambda^{+}_{\min}(A(q)) > 2. \tag{17}$$

Equivalently, every nonzero relaxation rate in (7) *is larger than $2k_0$.*

Proof. Relabel the coordinates so that $q_1 \geq q_2 \geq \cdots \geq q_N$, and set $d_i = 1/q_i$, so $d_1 \leq \cdots \leq d_N$. Let the eigenvalues of $A(q)$ be $0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_N$. Rank-one interlacing gives

$$0 = \lambda_1 \leq d_1 \leq \lambda_2 \leq d_2 \leq \cdots . \tag{18}$$

If $q_1 < 1/2$, then $d_1 > 2$, and (18) immediately gives $\lambda_2 > 2$.

If $q_1 = 1/2$, interlacing gives $\lambda_2 \geq 2$. The value 2 cannot be an eigenvalue. Indeed, $A(q)y = 2y$ would imply

$$(d_i - 2)y_i = c, \qquad c = \mathbf{1}^{\mathsf{T}} y. \tag{19}$$

The first coordinate gives $c = 0$. Since $N \geq 3$ and all q_i are positive, $q_i < 1/2$ for $i \geq 2$, so (19) gives $y_i = 0$ for those indices. It then also gives $y_1 = c = 0$, excluding a nonzero eigenvector.

It remains to consider $q_1 > 1/2$. This largest coordinate is unique. Put $s = 1 - q_1 \in (0, 1/2)$; then $d_1 < 2 < d_2$. On (d_1, d_2) the determinant lemma gives the secular function

$$F(\lambda) = 1 - \sum_{i=1}^{N} \frac{1}{d_i - \lambda}, \qquad F'(\lambda) = -\sum_{i=1}^{N} \frac{1}{(d_i - \lambda)^2} < 0. \tag{20}$$

It decreases from $+\infty$ to $-\infty$, and its unique zero is λ_2.

For $g(t) = t/(1 - 2t)$ on $[0, 1/2)$, direct algebra gives

$$g(a + b) - g(a) - g(b) = \frac{4ab(1 - a - b)}{(1 - 2a)(1 - 2b)(1 - 2a - 2b)} > 0 \tag{21}$$

whenever $a, b > 0$ and $a + b < 1/2$. The $N - 1 \geq 2$ positive numbers $q_2, \ldots, q_N$ sum to s, so repeated strict superadditivity yields $\sum_{i \geq 2} g(q_i) < g(s)$. Hence

$$\sum_{i=1}^{N} \frac{1}{d_i - 2} = -\frac{1 - s}{1 - 2s} + \sum_{i \geq 2} g(q_i) \tag{22}$$

$$< -\frac{1 - s}{1 - 2s} + \frac{s}{1 - 2s} = -1. \tag{23}$$

Thus $F(2) > 0$. Since F is strictly decreasing and vanishes at λ_2, we conclude that $\lambda_2 > 2$. The three cases exhaust the positive simplex. □

The strictness in (17) depends on both positivity of every q_i and $N \geq 3$. It is therefore a finite-offset statement; it does not assert a uniform positive margin above 2 as the boundary of the simplex is approached.

4 Phase bound, equality, and optimality

Corollary 4 (Weak-response phase bound)**.** *For every finite-offset model* (2) *with $N \geq 3$ and every finite real Fourier drive,*

$$A_2 - 2k_0 B_2 = \frac{1}{2} \sum_{n,\mu > 0} \frac{\omega_n^2 \lambda_\mu (\lambda_\mu - 2k_0) |a_{n\mu}|^2}{\lambda_\mu^2 + \omega_n^2} \geq 0. \tag{24}$$

If at least one time-dependent energy difference is nonconstant, then $A_2 > 2k_0 B_2 > 0$. Equality holds exactly when all positive-frequency Fourier coefficients are common to every state, in which case $A_2 = B_2 = 0$.

Proof. Equation (24) follows by subtracting twice k_0 times (12) from (11). Every nonzero factor $\lambda_\mu - 2k_0$ is strictly positive by Theorem 3. Moreover, all $a_{n\mu}$ vanish if and only if $c_n - \mathbf{1}\pi^\mathsf{T} c_n = 0$ for every $n \geq 1$, which is precisely a common time-dependent energy shift. Such a shift leaves every rate invariant. Arbitrary zero-frequency shifts likewise make no contribution to the cycle averages. □

In the phase convention of Ref. [1], a nonzero weak response obeys

$$\tan \Delta\varphi = \Omega \frac{B_2}{A_2} < \frac{\Omega}{2k_0}. \tag{25}$$

Thus every response-active finite-offset $N \geq 3$ system lies strictly below the symmetric two-state ceiling. The strict inequality does not mean that the constant can be improved uniformly.

Proposition 5 (Sharpness)**.** *For each fixed $N \geq 3$, the constant 2 in (1) is the optimal uniform constant over response-active systems with finite offsets, but it is not attained.*

Proof. For $0 < \delta \leq 1$, take

$$x_1 = x_2 = 1, \qquad x_3 = \cdots = x_N = \delta, \qquad S = 2 + (N-2)\delta. \tag{26}$$

The vector $v = (1, -1, 0, \ldots, 0)^\mathsf{T}$ is orthogonal to $\mathbf{1}$ and is an exact eigenvector of (7) with eigenvalue $k_0 S$. The two smallest diagonal entries of $S\,\mathrm{diag}(1/x_i)$ are both S; interlacing therefore shows that this is the smallest positive eigenvalue. A single real harmonic proportional to v is centered because $\pi_1 = \pi_2$, and it excites only this mode. Consequently,

$$\frac{A_2}{B_2} = k_0 \left[2 + (N-2)\delta\right]. \tag{27}$$

Every $\delta > 0$ corresponds to finite energies. As $\delta \downarrow 0$, the energies of states $3, \ldots, N$ diverge and (27) approaches $2k_0$ from above. Hence any larger uniform constant fails for sufficiently small positive δ, whereas Theorem 3 excludes attainment at a finite offset. □

5 Discussion and limitations

The response formulas separate the physical and spectral parts of the result. The drive, frequency, and observable determine nonnegative modal weights, whereas the offsets determine the relaxation spectrum. Complete connectivity with a common attempt rate forces every response-active mode above $2k_0$. The sharp family shows that the slowest allowed mechanism is not a genuinely multi-state interior point: it is approached when only two equally weighted states remain dynamically relevant.

The nonuniform gap is not obtained from the ordinary unweighted complete-graph spectrum alone. When $\max_i q_i \leq 1/2$, interlacing almost settles the question; when one weight is dominant, the sign of the first secular root is controlled by the strict superadditivity identity (21). This is the only regime in which the elementary diagonal lower bound is insufficient.

Several restrictions are essential. First, the statement concerns only the coefficients defined by the limit $\varepsilon \to 0$; it is not a finite-amplitude theorem. Second, all edges have the same prefactor k_0 and use symmetric splitting. Third, the graph is complete and all offsets are finite. Missing links, edge-dependent prefactors, asymmetric splitting, transient states, and singular offsets are outside the proof. In particular, the broader multi-state conjectures in Ref. [1] remain unresolved.

6 Conclusion

We have proved the weak-driving inequality $A_2 \geq 2k_0B_2$ for every finite-offset symmetric complete N-state Markov network with $N \geq 3$. The exact spectral reduction yields more than the non-strict bound: a response-active interior system is always strict, equality is exactly the zero-response gauge class, and the constant is an unattained sharp infimum. The result turns the two-state phase scale into a precise limiting benchmark for this complete multi-state model while leaving finite-amplitude and heterogeneous-network extensions as distinct open problems.

Data and code availability

No experimental data were generated. The analytic result is supported by accompanying Python programs that (i) reconstruct the weak-response formula in both the probability and detailed-balance bases, (ii) perform exact-rational finite-instance gap checks, and (iii) verify the secular and superadditivity identities symbolically. These programs are verification aids and do not replace the proof. A repository location and persistent identifier should be added by the author before submission.

Acknowledgments and generative-AI disclosure

OpenAI Codex using model `gpt-5.6-sol` assisted with research ideation, algebraic calculations, verification code, literature discovery, and manuscript drafting. All scientific claims, references, calculations, and wording require the named author's independent review and approval before submission. No AI system is listed as an author.

Conflict of interest

The author must confirm the conflict-of-interest declaration before submission.

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